

GJ 8999 b Exoplanet Characterization

Introduction

In this assignment, we will estimate the mass and radius of the exoplanet GJ 8999 b, which is orbiting the M dwarf star GJ 8999.

Star Properties

The known properties of the star GJ 8999 are listed below:

- **Mass of GJ 8999 (M_{star}):** $0.2M_{\odot}$
- **Radius of GJ 8999 (R_{star}):** $0.2R_{\odot}$

1. Inclination of GJ 8999 b

The inclination (i) can be calculated using the following formula:

$$i = \arccos(R_{\text{star}} / R_{\text{planet}})$$

1.1 Transit Depth

The transit depth (Transit Depth) is given by the following expression:

$$\text{Transit Depth} = (R_{\text{planet}} / R_{\text{star}})^2$$

From the transit depth, the radius of the planet (R_{planet}) can be estimated as:

$$R_{\text{planet}} = R_{\text{star}} \sqrt{\text{Transit Depth}}$$

2. Radial Velocity Data

The mass of the planet (M_{planet}) can be derived from radial velocity data using the equation:

$$M_{\text{planet}} = \frac{K \sqrt{P}}{G \sin(i)}$$

$$\sin(i)$$

Or, more commonly expressed as:

$$M_{\text{planet}} = \frac{K \sqrt{P}}{G \sin^3(i)}$$

Where:

- K : Radial velocity amplitude
- P : Orbital period
- G : Universal gravitational constant

Conclusion

Using the above formulas and data, the mass and radius of GJ 8999 b will be calculated, and the results will be analyzed.